

Vector Fields and Line Integrals



Vector fields and divergence

- 1 For the vector field $\mathbf{F}(x, y) = \langle x^2y, xy^2 \rangle$, find the divergence $\nabla \cdot \mathbf{F}$.
 - 2 For $\mathbf{F}(x, y, z) = \langle y, -x, z^2 \rangle$, find the curl $\nabla \times \mathbf{F}$.
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Line integrals

- 3 Evaluate the line integral $\int_C (x + y) ds$ where C is the line segment from $(0, 0)$ to $(3, 4)$.
 - 4 Evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = \langle y, x \rangle$ along $\mathbf{r}(t) = \langle t, t^2 \rangle$, $0 \leq t \leq 1$.
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Conservative vector fields and potential functions

- 5 Determine whether $\mathbf{F}(x, y) = \langle 2xy, x^2 + 3y^2 \rangle$ is conservative, using the test $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$.
 - 6 Find a potential function $f(x, y)$ such that $\nabla f = \mathbf{F}$ for the field above.
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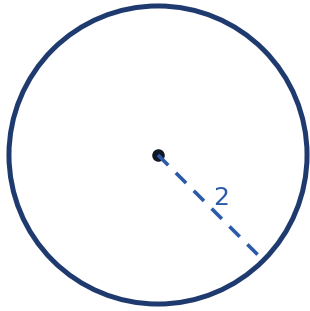
The Fundamental Theorem for line integrals

- 7 Using the potential function $f(x, y) = x^2y + y^3$ found above, evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along any path from $(0, 0)$ to $(1, 2)$, using the Fundamental Theorem for line integrals.
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Green's Theorem

- 8 Use Green's Theorem, $\oint_C P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$, to evaluate $\oint_C xy dx + x^2 dy$ where C is the boundary of the square $0 \leq x \leq 1$, $0 \leq y \leq 1$.
- 9 Use Green's Theorem to find the area of the region enclosed by a curve C , using the formula $A = \frac{1}{2} \oint_C (x dy - y dx)$, for the circle $x = 2 \cos t$, $y = 2 \sin t$, $0 \leq t \leq 2\pi$.

- 10 Use Green's Theorem to evaluate $\oint_C (3y) dx + (5x) dy$ where C is the circle $x^2 + y^2 = 4$, traversed counterclockwise.



Work done by a force field

- 11 A force field $\mathbf{F}(x, y) = \langle y, x \rangle$ acts on an object moving along the straight path from $(0, 0)$ to $(2, 0)$ then from $(2, 0)$ to $(2, 3)$. Find the total work done, $W = \int_C \mathbf{F} \cdot d\mathbf{r}$, by evaluating each leg separately.
- 12 Confirm the work found above using the potential function $f(x, y) = xy$ (since $\mathbf{F} = \langle y, x \rangle = \nabla f$) and the Fundamental Theorem for line integrals.

Testing for conservative fields

- 13 Determine whether $\mathbf{F}(x, y) = \langle y^2, 2xy + 1 \rangle$ is conservative, and if so, find a potential function.
- 14 Determine whether $\mathbf{F}(x, y) = \langle 3x^2y, x^3 + 2y \rangle$ is conservative, and find a potential function if so.

Applying Green's Theorem to circulation

- 15 Use Green's Theorem to evaluate the circulation $\oint_C (-y) dx + x dy$ around the boundary of the rectangle $0 \leq x \leq 3$, $0 \leq y \leq 2$, traversed counterclockwise.

Line integrals over piecewise paths

- 16 Evaluate $\int_C y dx + x dy$ where C consists of the segment from $(0, 0)$ to $(1, 0)$ followed by the segment from $(1, 0)$ to $(1, 1)$.

- 17 Confirm the result above using the potential function $f(x, y) = xy$ (since $\langle y, x \rangle = \nabla f$) and the endpoints $(0, 0)$ and $(1, 1)$.
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Flux and Green's Theorem (normal form)

- 18 Using the divergence form of Green's Theorem, $\oint_C \mathbf{F} \cdot \mathbf{n} \, ds = \iint_D \nabla \cdot \mathbf{F} \, dA$, find the outward flux of $\mathbf{F}(x, y) = \langle x, y \rangle$ across the boundary of the disk $x^2 + y^2 \leq 4$.
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A capstone conservative-field application

- 19 A particle moves through a force field $\mathbf{F}(x, y) = \langle 2xy^3, 3x^2y^2 + 4 \rangle$ from the point $(0, 0)$ to $(2, 1)$ along any path. First verify that $\mathbf{F}$ is conservative, then find a potential function $f(x, y)$, and finally use it to compute the total work done by the field on the particle.
- 20 For the vector field $\mathbf{F}(x, y, z) = \langle yz, xz, xy + 2z \rangle$, complete each step:
- a Find the divergence $\nabla \cdot \mathbf{F}$.
 - b Find the curl $\nabla \times \mathbf{F}$.
 - c Based on the curl, determine whether $\mathbf{F}$ could be conservative in 3D.